Brian Naklycky

Software Engineer

(609) 356-4306
Austin, TX
brian-nak1@hotmail.com

brianakl.github.io/
github.com/brianakl
linkedin.com/in/briannaklycky

Skills

Programming Languages Fluent: Python, C++, C, SQL | Intermediate: Java, MATLAB, Javascript, Go(Golang)
Tools & Libraries Unix/Linux, Jenkins, gdb debugger, pdb debugger,GitHub, Vim, Snowflake, AWS, PostgreSQL

Work Experience

Graduate Teaching Assistant | Natural Language Processing **Aug 2024 — Present**
University of Texas at Austin *Austin, TX*
- Assisted in teaching a graduate-level Natural Language Processing course, covering foundational models to modern machine learning and AI architectures
- Provided individualized support to students, demonstrating strong communication skills
- Developed proficiency in Neural Networks, Word Embeddings, Transformers, LLM factuality, and Dataset Artifact Analysis, while also gaining hands-on experience with database management systems such as PostgreSQL through a Unix/Linux enviornment.

Data Engineer Intern **May 2024 — Aug 2024**
Toyota Financial Services *Plano, TX*
- Reduced time-to-delivery for new data pipelines from thousands of hours to minutes, delivering a 99.9% increase in efficiency
- Developed a Python script that streamlined pipeline generation and metadata cleaning for thousands of individual tables across various ingestion and consumption pipelines
- Crafted SQL queries in Snowflake to create precise lists of table names for seamless data transfer, ensuring data accuracy& integrity
- Leveraged a range of technologies, including Python, Snowflake, SQL, Unix/Linux, GitHub, Jenkins, PowerBI, Jira, and AWS, to design, develop, and deploy scalable data pipelines in a CI/CD environment

Research Assistant | Quantum Network Communication **Nov 2022 — Aug 2022**
University of South Florida College of Computer Science & Engineering *Tampa, FL*
- Conducted research on vulnerabilities in healthcare network infrastructures to find use cases for Quantum network communication
- Reviewed and synthesized findings from leading papers on quantum computing and networking, integrating insights into potential machine learning applications for data security
- Created realistic simulations of classical and quantum networks using NS3 C++ library and SeQuEnCe Python network simulators on a Red Hat Linux compute cluster
- Experimented with a quantum TCP/IP and UDP/IP algorithm, comparing performance to a quantum acknowledgment channel in a busy network

Web Developer **Nov 2021 — Dec 2021**
The University of South Florida Quantum Initiative *Tampa, FL*
- Developed and designed a responsive, multi-device compatible website using HTML, CSS, and JavaScript to deploy on a Unix/Linux environment, demonstrating proficiency in front-end technologies
- Implemented ADA compliance and search engine optimization techniques, showcasing attention to accessibility and UX
- Gained hands-on experience in web application development within an academic research environment, aligning with the organizations focus on quantum computing
- Website URL: https://quantum.usf.edu/

Projects

Novel Transfer Learning for Efficient Large Language Model Training **Aug 2024 — Dec 2024**
Research Paper *Austin, TX*
- Developed BUS (Bootstrapped Model UpScaling), a novel method of combining the nGPT architecture with fractional HyperScaling for more efficient Large Language Model Training
- Conducted comprehensive comparisons across model sizes ranging from 10M to 100M parameters, demonstrating consistent performance gains and efficient scaling properties
- Investigated the impact of transfer learning on the efficient compute boundary, exploring opportunities to optimize computational resources while maintaining high accuracy.
- Conducting pretraining on a 100M parameter LLM running on a large GPU cluster
- Implemented the entire pipeline using Python, PyTorch, and HuggingFace Transformers, with extensive use of Numpy, Scipy, and Pandas for data analysis and visualization, as well as utilizing CUDA for accelerated training

Traveling Salesman Problem (TSP) **June 2021 — July 2021**
University of South Florida *Tampa, FL*
- Developed a comprehensive suite of 6 algorithms to tackle TSP, a classic NP-hard optimization challenge
- Implemented the entire project in C++, leveraging the Standard Template Library (STL) and Boost Graph Library to ensure efficient data structures and graph operations while gaining experience with the gdb debugging tool.
- Designed the algorithms to effectively solve TSP instances with moderately sized graphs, balancing performance and scalability.
- Demonstrated proficiency in advanced algorithm design, optimization techniques, and C++ programming, while gaining practical experience in solving real-world combinatorial optimization problems

Education

University of Texas at Austin Master’s in Computer Science Dec 2024
University of South Florida Bachelor’s in Computer Science Dec 2022